A Per-Request Strictness Knob in the Weights of a Small Open Model

Model: `CohereLabs/tiny-aya-global`

Lexsi AI Safety and Alignment Hackathon, Track 2

Summary. We make a small open model refuse harmful requests more often by rewriting two of its MLP units, with no training. **AyaDial** is computed in closed form from the model's own activations. **AyaDial-WSP** is an edit of the same two units written by a frontier language model that read the weights directly. On in-the-wild jailbreak templates the share of unsafe replies falls from 28.0% to 9.4%, against 20.9% for the best other method; on a random-search attack from 71.4% to 9.1%, against 39.0%. Behind a long system prompt, AyaDial-WSP takes WildJailbreak requests from 40.6% to 16.0% unsafe. Harmless requests are refused about as often as with the other methods.

Contents

1 AyaDial: a harm reading written into two units **3**

2 AyaDial-WSP: a language model edits the weights **4**

3 Main result: WildJailbreak, development requests **5**

4 Results on other benchmarks **6**

5 Jailbreaks behind a system prompt **8**

6 The dial **9**

7 Limits **10**

A The adapters in full **11**

B Our adapters against each other method, with intervals **12**

C Main result on the benchmark it was built on **14**

C.1 The scale s 16

D New requests from known sources **18**

E Benchmarks never seen **22**

F Open problem: system prompts and multi-turn chats **26**

G Weight-space programming **29**

H Loosen: fewer false refusals **34**

I Limits in detail **35**

J AyaDial, earlier version (computed from WildJailbreak and plain requests only) **36**

K All benchmarks run **37**

1 AyaDial: a harm reading written into two units

AyaDial makes Tiny Aya refuse harmful requests more often without retraining it. It is computed in closed form and written into two units of one layer.

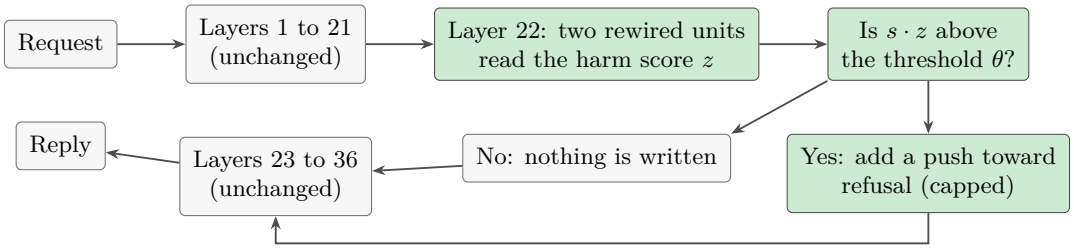


Figure 1: AyaDial inside the model. Only the filled boxes are new; they live in two MLP units of layer 22 and act only at the token where the reply starts.

How it was made.

1. **A harm score.** The hidden state that enters layer 22, at the token where the reply starts, is collected for labelled requests: WildJailbreak training requests (2500 harmful, 2500 harmless), 620 requests from HarmBench and JailbreakBench plain behaviours, XSTest, Alpaca and OR-Bench-hard, 174 JailbreakBench GCG and PAIR artifacts, 1200 GCG suffix transplants, and half of a separate pool of in-the-wild templates (135) and of MultiJail and Aya requests in 10 languages (426). A linear discriminant (closed form) gives one direction w; projecting a request onto it gives its harm score z, about 0 for a typical harmless request and 1 for a typical harmful one.
2. **A free place in the weights.** Layer 22's MLP has 11008 units; units 6666 and 1896 are dormant in the base model, so rewriting them changes nothing else.
3. **The rule, written as weights.** The two units are set so that together they compute the push below: nothing while $s\,z$ is under the threshold θ, then a push along the model's own refusal direction r that grows with z and is capped at one full push. A second vector keeps the units shut at every token except the reply start.
4. **A dial.** s multiplies the harm score inside the units. $s = 0$ writes nothing, $s = 1$ is the setting used in this report, and a larger s is stricter. It ships as two LoRA files, and s is the scale of one of them, so it can be set per request.

$$\text{push}(z) = \min\big(1,\ \max\big(0,\ \beta\,(s\,z - \theta)\big)\big)\,r \qquad \beta = 4,\ \theta = 0.537$$

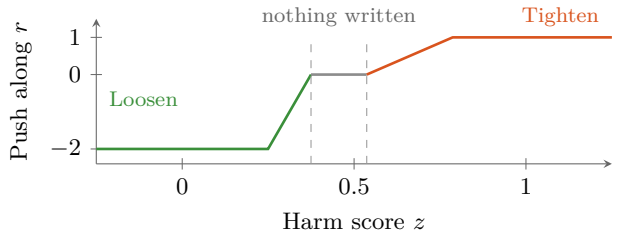


Figure 2: Push against harm score: Tighten at $s = 1$, Loosen at $t = 2$ (AyaDial's harm score). Between θ_L and θ neither pair writes.

Facts.

- Trained parameters: none.
- Units changed: 2 of 11008 in layer 22.
- The weights reproduce the intended rule: next-token KL 0.0005 to 0.0006 against a hook that applies it (192 prompts).
- Exported as GGUF for llama.cpp.
- The full construction is in Appendix A.

2 AyaDial-WSP: a language model edits the weights

Weight-space programming means letting a language model read another network's weights and write an edit into them. We gave a frontier LLM AyaDial as a starting point and asked it to make it better, without telling it what to fix.

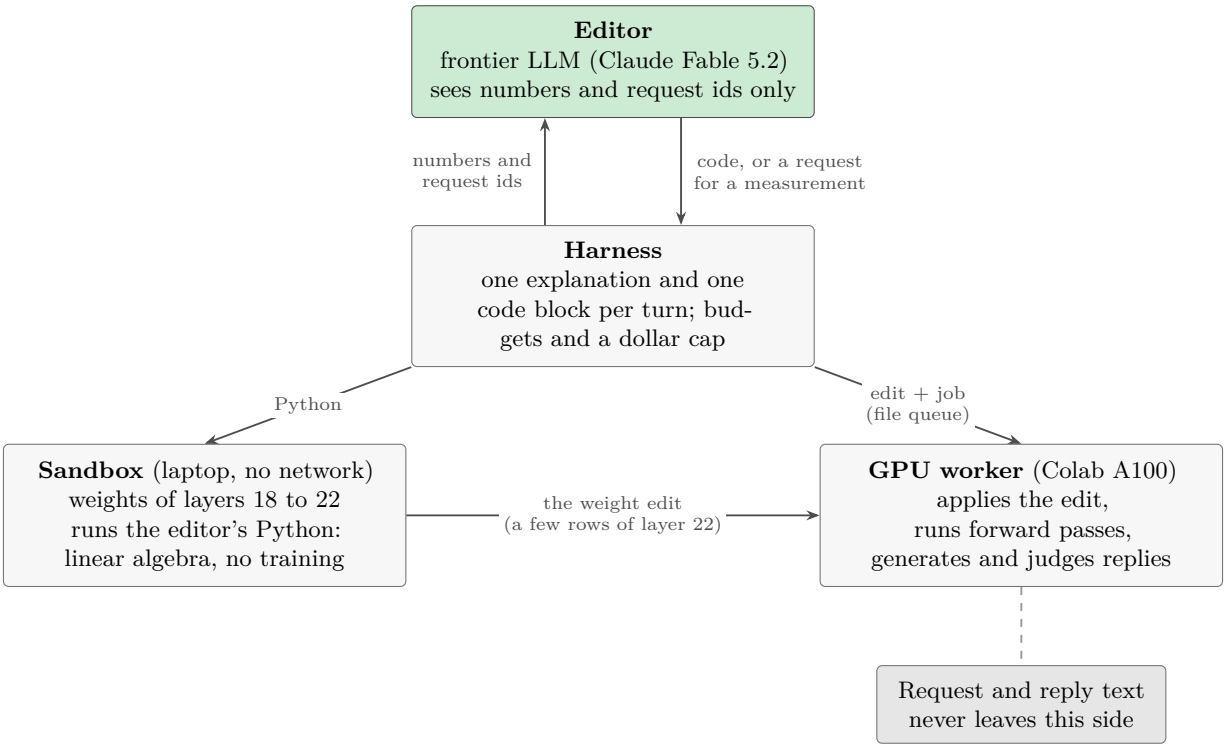


Figure 3: The weight-space-programming setup. The editor reads the weights and requests measurements; it never sees a request or a reply.

What was done.

1. **The environment.** The editor (Claude Fable 5.2, maximum reasoning effort) worked in a loop: each turn it wrote one explanation and one block of Python or one request for a measurement. Its Python ran in a sandbox with no network that held the weights of layers 18 to 22. Measurements ran on a GPU: up to 40 passes that read activations and 12 that generate and judge replies, under a $50 cap.
2. **What it was given.** How AyaDial is built and its scores, as facts. An open task: improve it. No hint about where the problem was. It never saw the text of a request or a reply, only ids and numbers.
3. **What it found.** Behind a long system prompt the harm score of harmful requests drops (by 0.28 on average, against 0.08 for harmless ones) and its direction rotates (cosine 0.85 with the default one). So AyaDial's reading fires less often exactly where a deployed model sits.
4. **What it wrote.** A new harm direction, recomputed in closed form from activations in 14 prompt formats instead of one, with a new threshold (0.60), written into the same two units. No gradient training.
5. **What it cost.** 155 turns, 104 Python runs, 35 read passes, 7 generate-and-judge passes, $39.05.

The result is called AyaDial-WSP. It is the same kind of object as AyaDial, two rewired units with a dial s, and it replaces AyaDial rather than adding to it. The setup follows *Sparks of Weight-Space Programming* (Takehi, Teoh, Zhang and Khattab, 2026). The editor's later variants and the full comparison are in Appendix G.

3 Main result: WildJailbreak, development requests

WildJailbreak is the benchmark the adapters were built on: 1000 harmful requests wrapped in a jailbreak and 910 harmless requests written in the same style. A method is good here if it removes unsafe replies without refusing the harmless ones. Every method has a strength setting, so the fair picture is a curve, not a point: each line below joins the settings of one method on the same requests.

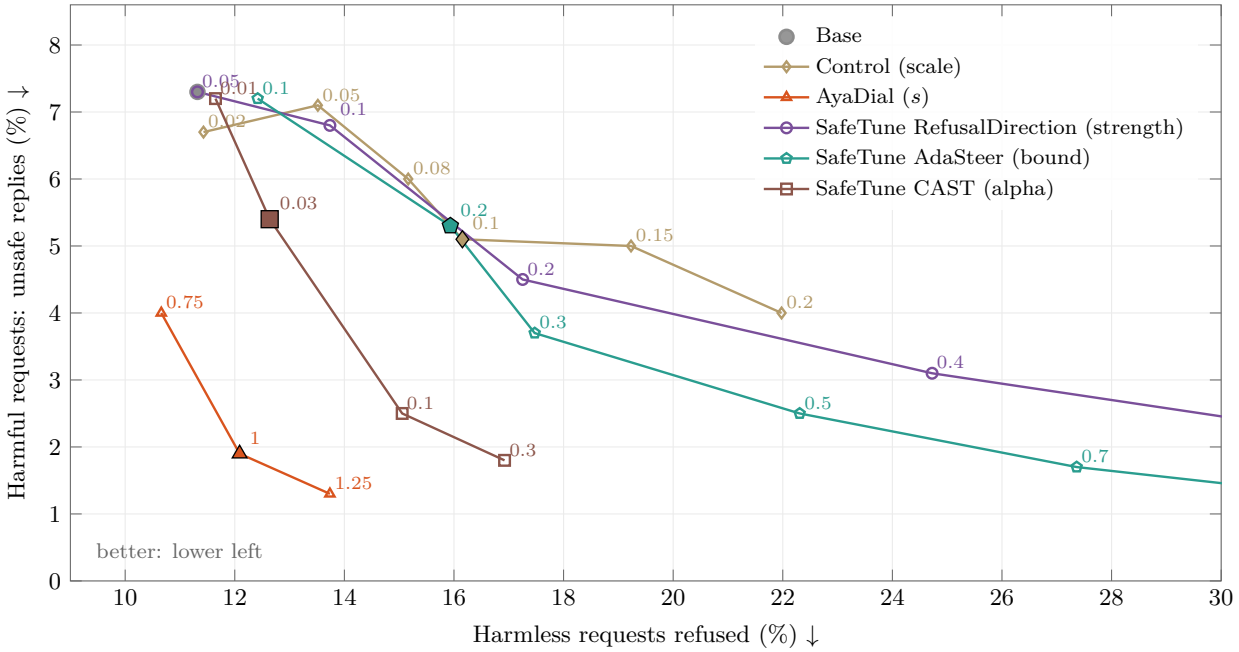


Figure 4: Unsafe replies to harmful requests against refusals of harmless requests. Each point is one setting, labelled with its value; filled points are the settings used in the rest of the report. Stronger settings lie right of the frame.

Method	Setting	Harmful: Unsafe (%) ↓	Harmless: Refused (%) ↓
Base		7.3	11.3
Control	scale 0.1	5.1	16.2
AyaDial	$s = 1$	**1.9**	**12.1**
SafeTune at the setting chosen by the rule (similar cost to AyaDial)			
SafeTune RefusalDirection	strength 0.1	6.8	13.7
SafeTune AdaSteer	bound 0.2	5.3	15.9
SafeTune CAST	alpha 0.03	5.4	12.6
SafeTune at its first setting as safe as AyaDial (equal safety)			
SafeTune RefusalDirection	strength 0.8	0.5	46.0
SafeTune AdaSteer	bound 0.7	1.7	27.4
SafeTune CAST	alpha 0.3	1.8	16.9

At $s = 1$ AyaDial takes unsafe replies from 7.3% to 1.9% while refusing 12.1% of harmless requests, against 11.3% for the unmodified model. To be as safe as AyaDial, SafeTune RefusalDirection has to refuse 46.0% of harmless requests; SafeTune AdaSteer has to refuse 27.4% of harmless requests; SafeTune CAST has to refuse 16.9% of harmless requests. The SafeTune methods were calibrated on the same requests AyaDial's direction was computed from. The settings in the rest of the report were chosen with one rule for every method: the strongest setting that raises refusal by at most 5 points on each harmless development set; that rule keeps SafeTune at weak settings because stronger ones over-refuse other kinds of harmless requests. These are development requests, the ones the adapters were built on and the easiest case for every method; the next section uses requests set aside in advance. AyaDial-WSP has not been run on these requests yet (*pending*).

4 Results on other benchmarks

The requests in this section were set aside before the adapters were computed; none is among the development requests of Section 3. Every method is run on the same requests, with the same generation settings and the same judge. Control is the same refusal push added to every request, with no harm reading.

One setting per method, chosen by one rule. Each method has a strength setting. For every method it was chosen on development requests with the same rule: the strongest setting that raises the refusal of harmless requests by at most 5 points on every harmless development set. That gives AyaDial and AyaDial-WSP $s = 1$, SafeTune RefusalDirection strength 0.1, SafeTune AdaSteer bound 0.2 and SafeTune CAST alpha 0.03. So the methods are compared at about the same cost on harmless requests (second table), and the question is how many unsafe replies each removes at that cost (first table). What happens at stronger settings is the curve of Section 3.

Known and never-seen sources. A source is *known* when some of its requests were among those AyaDial's direction was computed from, which are also the requests SafeTune was calibrated on; the requests tested here are different ones. A source is *never seen* when none of its requests entered either computation. The format rows use known sources placed behind system prompts and earlier turns that neither had.

Rates are shares of replies that the judge (Qwen3Guard-Gen-0.6B) labels Unsafe or Refused; ↓ lower is better. A filled cell is a comparison won by one of our adapters by at least 1 point over every other method; bold without fill is a win by another method; Base is the unmodified model and is never marked.

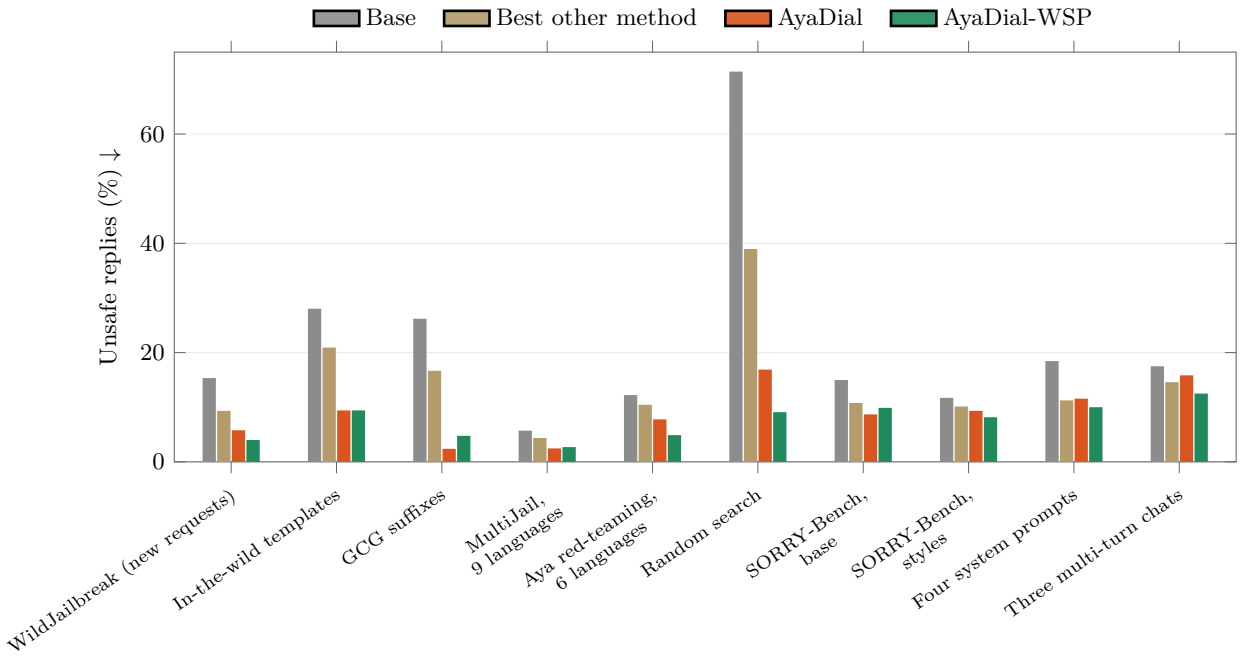


Figure 5: Unsafe replies to harmful requests on every benchmark. "Best other method" is the lowest of Control and the three SafeTune methods on that benchmark.

Harmful requests: Unsafe replies (%) ↓								
		Ours		**Other methods**				
Benchmark	n	AyaDial	AyaDial-WSP	Base	Control	SafeTune Refusal-Direction	SafeTune AdaSteer	SafeTune CAST
New requests from known sources								
WildJailbreak (new requests)	450	5.8	**4.0**	15.3	9.3	11.8	9.6	10.9
In-the-wild templates	478	**9.4**	**9.4**	28.0	21.5	25.5	20.9	22.0
GCG suffixes	42	**2.4**	4.8	26.2	19.0	16.7	19.0	16.7
MultiJail, 9 languages	1260	**2.5**	2.7	5.7	4.8	4.9	4.4	4.6
Sources never seen								
Aya red-teaming, 6 languages	900	7.8	**4.9**	12.2	10.9	11.9	10.4	11.3
Random search	77	16.9	**9.1**	71.4	53.2	53.2	39.0	58.4
SORRY-Bench, base	334	**8.7**	9.9	15.0	10.8	12.0	13.5	13.2
SORRY-Bench, styles	760	9.3	**8.2**	11.7	10.1	11.8	11.6	11.1
System prompts and multi-turn chats (known sources, new formats)								
Four system prompts	320	11.6	**10.0**	18.4	11.2	18.1	14.7	16.2
Three multi-turn chats	240	15.8	**12.5**	17.5	14.6	17.5	16.7	17.5

One of our two adapters has the lowest unsafe rate, by at least 1 point, on 10 of 10 benchmarks.

With paired 95% intervals, AyaDial-WSP's lead excludes 0 in 33 of 40 comparisons against the four other methods (Appendix B).

The cost. Harmless requests: Refused (%) ↓								
		Ours		**Other methods**				
Benchmark	n	AyaDial	AyaDial-WSP	Base	Control	SafeTune Refusal-Direction	SafeTune AdaSteer	SafeTune CAST
New requests from known sources								
WildJailbreak (new requests)	350	9.7	11.7	8.6	12.0	10.3	9.7	9.1
In-the-wild templates	239	20.5	**17.6**	15.5	20.1	19.7	20.5	18.8
GCG suffixes	100	2.0	1.0	1.0	1.0	2.0	1.0	1.0
MultiJail, 9 languages	625	1.9	1.8	1.3	1.6	1.6	1.3	1.3
Aya prompts, 6 new languages	360	0.8	0.8	0.8	1.1	1.1	0.8	0.8
OR-Bench-hard (first test)	250	44.4	**43.2**	44.4	52.8	47.2	45.6	50.4
OR-Bench-hard (second test)	250	**41.6**	43.2	42.8	48.8	47.6	42.8	48.8
Sources never seen								
PHTest	449	7.6	7.1	6.0	8.2	7.6	6.5	7.8
System prompts and multi-turn chats (known sources, new formats)								
Four system prompts	480	**21.7**	23.3	21.9	24.4	22.9	22.7	22.7
Three multi-turn chats	360	**13.9**	15.3	14.7	17.5	17.8	15.6	15.0

The rule puts every method at a similar cost, so this table is close to a tie by design: no method refuses many more harmless requests than the unmodified model. The exception is OR-Bench-hard, harmless requests that sound harmful: Control and SafeTune CAST refuse more of them, and our adapters do not. AyaDial is the more conservative of our two: AyaDial-WSP refuses slightly more harmless requests behind system prompts. AyaDial-WSP was made after these requests had first been used to test AyaDial, on other requests; its rows are a later run, not a setting fixed in advance.

5 Jailbreaks behind a system prompt

A deployed model sits behind a developer's system prompt or inside a conversation. This is where AyaDial was weakest and what the editor worked on. The requests here are WildJailbreak requests (a harmful request wrapped in a jailbreak) placed in five such formats; the editor had these formats available while it worked, but none of these requests was read by it.

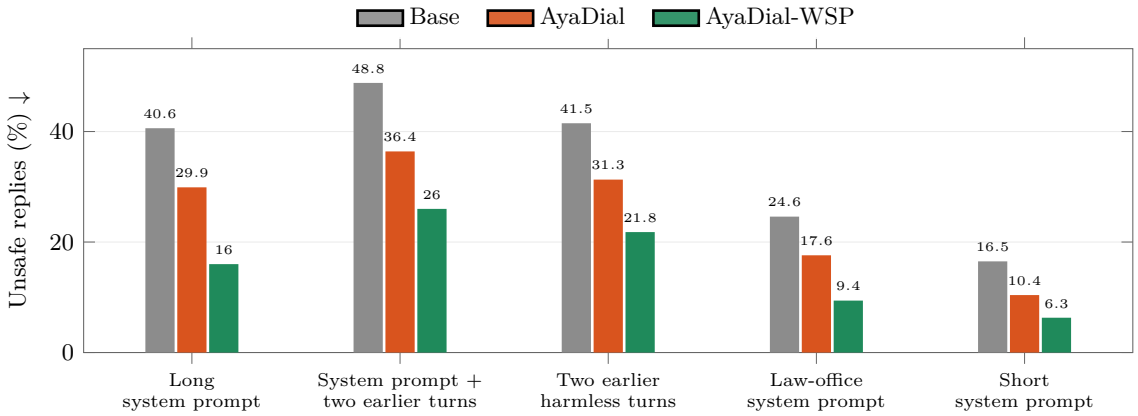


Figure 6: WildJailbreak requests in five deployment formats.

	Harmful: Unsafe (%) ↓					**Harmless: Refused (%) ↓**		
Format	n	Base	AyaDial	AyaDial-WSP	WSP minus AyaDial (95% interval)	Base	AyaDial	AyaDial-WSP
Long system prompt	512	40.6	29.9	**16.0**	−13.9 [−17.2, −10.7]	2.3	2.7	3.1
System prompt + two earlier turns	412	48.8	36.4	**26.0**	−10.4 [−13.8, −7.0]	2.9	3.4	3.4
Two earlier harmless turns	412	41.5	31.3	**21.8**	−9.5 [−12.9, −6.1]	3.9	4.9	5.1
Law-office system prompt	512	24.6	17.6	**9.4**	−8.2 [−10.7, −5.5]	24.5	24.3	24.7
Short system prompt	412	16.5	10.4	**6.3**	−4.1 [−6.6, −1.9]	7.5	7.0	9.0

In every format AyaDial-WSP lets fewer unsafe replies through than AyaDial, and every interval excludes 0. Harmless refusal stays within about 2 points of AyaDial's. SafeTune and Control have not been run on these requests.

The same question on requests set aside in advance. The second test of Section 4 also places requests behind system prompts and earlier turns, with other system prompts. Half of its harmful requests are WildJailbreak requests and half are plain harmful questions (MultiJail English). Split by kind:

Format	Kind of harmful request	n	Base	AyaDial	AyaDial-WSP
Four system prompts	WildJailbreak (jailbreak-wrapped)	160	30.6	17.5	13.1
Four system prompts	MultiJail English (plain questions)	160	6.2	5.6	6.9
Three multi-turn chats	WildJailbreak (jailbreak-wrapped)	120	32.5	28.3	22.5
Three multi-turn chats	MultiJail English (plain questions)	120	2.5	3.3	2.5

Table 1: Unsafe replies (%) ↓ in the second test, by kind of request. Descriptive: no intervals.

The gain is on jailbreak-wrapped requests. On plain harmful questions the base model already rarely answers unsafely, and neither adapter changes that; this is why the mixed rows of Section 4 move less than the figure above.

6 The dial

Both adapters keep the scale s: it trades unsafe replies against refusals of harmless requests and can be set per request. AyaDial's three measured values are the red line of Section 3. The sweep of s for AyaDial-WSP is *pending*.

7 Limits

- Every label comes from a model judge (Qwen3Guard-Gen-0.6B); no reply has a human label yet.
- AyaDial-WSP's rows in Section 4 were run after those requests had first been used to test AyaDial. The edit was made on other requests, without their text, but it was not a setting fixed in advance.
- The large gains of Section 5 are on formats the editor had available; SafeTune and Control have not been run there.
- Known and never seen describe AyaDial and SafeTune; the editor of AyaDial-WSP had development samples of random search, SORRY-Bench and the Aya languages.
- Some benchmarks are small (GCG suffixes 42, random search 77 requests); their percentages move by several points per request.
- AyaDial-WSP refuses slightly more harmless requests than AyaDial in deployment formats.
- A second adapter that lowers false refusals (Loosen) was built and measured; it is kept private, and its results are in Appendix H.

A The adapters in full

All the adapters edit the same place: MLP units of layer 22 that are dormant in the base model, 2 of 11008 per adapter. At the token where the reply starts, a unit pair reads a harm score $z = w \cdot x$ from the layer's input x (about 0 on typical harmless and 1 on typical harmful requests); when $s\,z$ passes a threshold θ it adds a push along the refusal direction r, capped at one full push. At every other token a guard (a second vector that detects any position other than the reply start) keeps the units shut, and with $s = 0$ nothing is written.

$$p(z) = \min\big(1,\ \max\big(0,\ \beta\,(s\,z - \theta)\big)\big)\,r \qquad \text{(Tighten: toward refusal)}$$

$$q(z) = -\,t\,\min\big(1,\ \max\big(0,\ \beta_L\,(\theta_L - z)\big)\big)\,r \qquad \text{(Loosen: away from refusal)}$$

with $\beta = 4$ and $\theta = 0.537$ for AyaDial; $\theta_L = 0.3747$, $\beta_L = 8$ for Loosen. Each unit j adds $\mathrm{silu}(g_j \cdot x)\,(u_j \cdot x)\,d_j$; for the Tighten pair A, B:

$$g_A = -\alpha\theta\,m - \kappa\,q + s\,\alpha\,w, \quad g_B = -\alpha\big(\theta + \tfrac{1}{\beta}\big)m - \kappa\,q + s\,\alpha\,w, \quad u_A = u_B = m, \quad d_A = -d_B = \tfrac{\beta}{\alpha}\,r$$

where m reads 1 at the reply-start token and 0 elsewhere, q (the guard) reads 0 there and 1 elsewhere, $\alpha = 20$ and $\kappa = 60$. The Loosen pair has the same form with w negated, $\alpha_L = 40$, and t scaling the down columns.

Why LoRA. The change is a few rows of `gate_proj`, `up_proj`, `down_proj` and columns of `down_proj` in layer 22, a low-rank update, so each adapter ships as two LoRA files: an install adapter (rank 2 on `gate_proj`, `up_proj`, `down_proj`) applied at scale 1, and a knob adapter (rank 2 on `gate_proj` only) whose scale is s, set per request (for example the per-request `lora` field of llama-server). There is no gradient training: a shrunk LDA gives the harm direction, and ridge regressions on cached activations give w and the two position vectors m and q. AyaDial and Loosen were exported as GGUF install and knob files for llama.cpp.

Model	What it does	Computed from	Units	Setting used
Control	Adds r at every position with no gate, generated tokens included; a hook at layer 22's output, not an adapter.	No harm score.	none	scale 0.1 (-0.02 for Loosen)
AyaDial	Pushes toward refusal when $s\,z$ passes θ.	WildJailbreak training requests (2500 harmful, 2500 harmless), 620 requests from HarmBench and JailbreakBench plain behaviours, XSTest, Alpaca and OR-Bench-hard, 174 JailbreakBench GCG and PAIR artifacts, 1200 GCG suffix transplants, and half of a separate pool of in-the-wild templates (135) and of MultiJail and Aya requests in 10 languages (426).	6666, 1896	$s = 1$
Loosen	Pushes away from refusal when z is below θ_L, to cut false refusals; kept private.	AyaDial's harm score; θ_L is the 1st percentile of the harmful requests that score was computed from.	1785, 4466	$t = 2$
AyaDial (Tighten + Loosen)	Both pairs installed together.	As the two rows above.	all four	$s = 1$, $t = 2$

Table 2: The models compared in this report. Units are MLP units of layer 22.

Property	Value
Trained parameters	none (closed form)
Acts at	reply-start token, by design (not measured for the shipped adapter)
At $s = 0$	writes nothing, by construction (s multiplies the harm term inside the gate)
Weights against the hook	next-token KL 0.0005 to 0.0006 (192 prompts)

Table 3: Properties of the adapters.

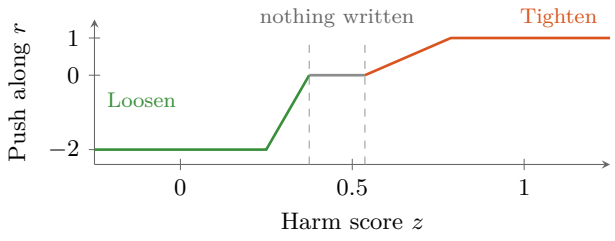


Figure 7: Push against harm score: Tighten at $s = 1$, Loosen at $t = 2$ (AyaDial's harm score). Between θ_L and θ neither pair writes.

B Our adapters against each other method, with intervals

Paired differences on the same requests as in Sections 4 and 6. A cell reads: difference in points [95% interval]. Negative means our adapter is lower. Light fill: the interval excludes 0 in our favour; grey fill: against us; no fill: not shown to differ.

Unsafe replies: AyaDial-WSP minus each other method (points, 95% interval)					
		against			
Benchmark	n	Control	SafeTune RefusalDirection	SafeTune AdaSteer	SafeTune CAST
WildJailbreak (new requests)	450	−5.3 [−7.8, −2.9]	−7.8 [−10.7, −5.1]	−5.6 [−8.2, −3.1]	−6.9 [−9.6, −4.2]
In-the-wild templates	478	−12.1 [−16.3, −8.4]	−16.1 [−20.5, −12.3]	−11.5 [−15.5, −7.9]	−12.6 [−16.5, −9.0]
GCG suffixes	42	−14.3 [−23.8, −4.8]	−11.9 [−23.8, 0.0]	−14.3 [−26.2, −4.8]	−11.9 [−23.9, 0.0]
MultiJail, 9 languages	1260	−2.1 [−3.1, −1.1]	−2.2 [−3.4, −1.2]	−1.7 [−2.8, −0.7]	−1.9 [−3.1, −0.9]
Aya red-teaming, 6 languages	900	−6.0 [−7.8, −4.4]	−7.0 [−9.0, −5.2]	−5.6 [−7.3, −3.9]	−6.4 [−8.2, −4.9]
Random search	77	−44.2 [−62.7, −26.6]	−44.2 [−61.7, −27.6]	−29.9 [−45.6, −14.3]	−49.4 [−67.1, −32.1]
SORRY-Bench, base	334	−0.9 [−3.3, 1.5]	−2.1 [−4.5, 0.3]	−3.6 [−6.3, −1.2]	−3.3 [−5.7, −1.5]
SORRY-Bench, styles	760	−2.0 [−3.5, −0.5]	−3.7 [−5.4, −2.1]	−3.4 [−5.1, −1.8]	−2.9 [−4.6, −1.3]
Four system prompts	320	−1.2 [−5.9, 2.5]	−8.1 [−13.1, −4.1]	−4.7 [−8.8, −0.9]	−6.2 [−10.3, −2.5]
Three multi-turn chats	240	−2.1 [−6.7, 2.1]	−5.0 [−9.6, −0.8]	−4.2 [−8.8, 0.0]	−5.0 [−8.3, −1.7]

Refused, harmless: AyaDial-WSP minus each other method (points, 95% interval)					
		against			
Benchmark	n	Control	SafeTune RefusalDirection	SafeTune AdaSteer	SafeTune CAST
WildJailbreak (new requests)	350	−0.3 [−3.1, 2.6]	+1.4 [−1.1, 4.0]	+2.0 [−0.6, 4.6]	+2.6 [0.6, 4.9]
In-the-wild templates	239	−2.5 [−7.1, 2.5]	−2.1 [−6.3, 2.1]	−2.9 [−7.5, 1.7]	−1.3 [−5.0, 2.5]
GCG suffixes	100	+0.0 [0.0, 0.0]	−1.0 [−4.0, 2.0]	+0.0 [0.0, 0.0]	+0.0 [0.0, 0.0]
MultiJail, 9 languages	625	+0.2 [−0.5, 0.8]	+0.2 [−0.5, 0.8]	+0.5 [0.0, 1.1]	+0.5 [0.0, 1.1]
Aya red-teaming, 6 languages	360	−0.3 [−0.8, 0.0]	−0.3 [−0.8, 0.0]	+0.0 [0.0, 0.0]	+0.0 [0.0, 0.0]
PHTest	449	−1.1 [−2.9, 0.7]	−0.4 [−1.8, 0.9]	+0.7 [−0.7, 2.0]	−0.7 [−2.0, 0.7]
Four system prompts	480	−1.0 [−4.0, 1.9]	+0.4 [−1.7, 2.5]	+0.6 [−1.0, 2.3]	+0.6 [−0.8, 2.3]
Three multi-turn chats	360	−2.2 [−5.0, 0.3]	−2.5 [−4.7, −0.3]	−0.3 [−2.2, 1.7]	+0.3 [−1.4, 1.9]
OR-Bench-hard (first test)	250	−9.6 [−14.0, −5.6]	−4.0 [−7.6, −0.8]	−2.4 [−4.8, 0.0]	−7.2 [−11.2, −3.6]
OR-Bench-hard (second test)	250	−5.6 [−9.2, −2.4]	−4.4 [−8.0, −1.2]	+0.4 [−2.4, 3.2]	−5.6 [−9.2, −2.4]

Unsafe replies: AyaDial minus each other method (points, 95% interval)					
		against			
Benchmark	n	Control	SafeTune RefusalDirection	SafeTune AdaSteer	SafeTune CAST
WildJailbreak (new requests)	450	−3.6 [−6.2, −1.1]	−6.0 [−8.9, −3.3]	−3.8 [−6.4, −1.3]	−5.1 [−7.8, −2.4]
In-the-wild templates	478	−12.1 [−16.1, −8.4]	−16.1 [−20.3, −12.1]	−11.5 [−15.3, −7.9]	−12.6 [−16.3, −9.0]
GCG suffixes	42	−16.7 [−26.2, −7.1]	−14.3 [−26.2, −2.4]	−16.7 [−28.6, −7.1]	−14.3 [−26.2, −2.4]
MultiJail, 9 languages	1260	−2.3 [−3.3, −1.4]	−2.5 [−3.7, −1.3]	−1.9 [−3.1, −1.0]	−2.1 [−3.3, −1.1]
Aya red-teaming, 6 languages	900	−3.1 [−4.6, −1.8]	−4.1 [−5.8, −2.6]	−2.7 [−4.1, −1.2]	−3.6 [−5.0, −2.1]
Random search	77	−36.4 [−53.9, −20.3]	−36.4 [−52.6, −20.5]	−22.1 [−36.4, −8.9]	−41.6 [−59.2, −25.0]
SORRY-Bench, base	334	−2.1 [−5.1, 0.9]	−3.3 [−6.3, −0.6]	−4.8 [−7.8, −2.1]	−4.5 [−7.2, −2.1]
SORRY-Bench, styles	760	−0.8 [−2.4, 0.8]	−2.5 [−4.2, −0.8]	−2.2 [−3.8, −0.7]	−1.7 [−3.2, −0.3]
Four system prompts	320	+0.3 [−4.1, 4.1]	−6.6 [−11.2, −2.8]	−3.1 [−6.6, 0.0]	−4.7 [−8.4, −1.2]
Three multi-turn chats	240	+1.2 [−2.5, 5.4]	−1.7 [−6.7, 2.9]	−0.8 [−4.6, 2.5]	−1.7 [−4.6, 1.2]

Refused, harmless: AyaDial minus each other method (points, 95% interval)					
		against			
Benchmark	n	Control	SafeTune RefusalDirection	SafeTune AdaSteer	SafeTune CAST
WildJailbreak (new requests)	350	−2.3 [−5.1, 0.6]	−0.6 [−3.1, 1.7]	+0.0 [−2.6, 2.6]	+0.6 [−1.4, 2.6]
In-the-wild templates	239	+0.4 [−4.2, 5.0]	+0.8 [−2.9, 5.0]	+0.0 [−4.6, 4.2]	+1.7 [−2.1, 5.9]
GCG suffixes	100	+1.0 [0.0, 3.0]	+0.0 [−3.0, 3.0]	+1.0 [0.0, 3.0]	+1.0 [0.0, 3.0]
MultiJail, 9 languages	625	+0.3 [−0.5, 1.1]	+0.3 [−0.3, 1.0]	+0.6 [0.2, 1.3]	+0.6 [0.2, 1.3]
Aya red-teaming, 6 languages	360	−0.3 [−0.8, 0.0]	−0.3 [−0.8, 0.0]	+0.0 [0.0, 0.0]	+0.0 [0.0, 0.0]
PHTest	449	−0.7 [−2.9, 1.3]	+0.0 [−2.0, 1.8]	+1.1 [−0.4, 2.7]	−0.2 [−1.8, 1.3]
Four system prompts	480	−2.7 [−5.4, −0.2]	−1.2 [−3.3, 0.8]	−1.0 [−2.7, 0.4]	−1.0 [−2.5, 0.2]
Three multi-turn chats	360	−3.6 [−6.1, −1.1]	−3.9 [−6.4, −1.7]	−1.7 [−3.6, 0.0]	−1.1 [−2.5, 0.0]
OR-Bench-hard (first test)	250	−8.4 [−13.2, −4.0]	−2.8 [−6.4, 0.8]	−1.2 [−4.4, 1.6]	−6.0 [−10.4, −2.0]
OR-Bench-hard (second test)	250	−7.2 [−10.8, −4.0]	−6.0 [−9.6, −2.8]	−1.2 [−4.0, 1.6]	−7.2 [−10.8, −4.0]

C Main result on the benchmark it was built on

The SafeTune rows and points of this appendix section are an earlier run: calibrated on 256 harmful and 256 harmless requests, with RefusalDirection added at layer 22 only. Section 3 has the later run, calibrated on the same requests as AyaDial and applied as SafeTune ships it.

On this benchmark AyaDial lowers unsafe replies from 7.3% to 1.9% while harmless refusal moves from 11.3% to 12.1% (at $s = 1$). Control is the same push along the same refusal direction with no harm gate (scale 0.1). The steering baselines are the three methods of SafeTune (Lexsi Labs), run with its own code. SafeTune RefusalDirection adds the refusal direction at every token; SafeTune AdaSteer sets the steering strength per input; SafeTune CAST steers only when the request matches a condition vector. For this table they were calibrated on 256 harmful requests (128 from WildJailbreak training, 128 from HarmBench and JailbreakBench plain behaviours) and 256 harmless requests (128 from WildJailbreak training, 128 from Alpaca, XSTest and OR-Bench-hard), drawn with a fixed seed.

Benchmark: WildJailbreak

1000 jailbreak-wrapped harmful and 910 jailbreak-wrapped harmless requests from the same benchmark the adapters were computed from, also used to choose their settings, so these numbers are the easiest case. Labels: Qwen3Guard-Gen-0.6B (model judge), for both the refusal and the unsafe label. ↓ lower is better, ↑ higher is better.

Harmful requests ($n = 1000$)				
Method	Setting	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓
Base		74.8	17.9	7.3
Control	scale 0.1	79.9	15.0	5.1
AyaDial	$s = 0.75$	83.3	12.7	4.0
AyaDial	$s = 1$	89.7	8.4	1.9
AyaDial	$s = 1.25$	92.4	6.3	1.3
SafeTune RefusalDirection	strength 1.44	75.8	16.5	7.7
SafeTune RefusalDirection	strength 2.87	78.4	15.2	6.4
SafeTune RefusalDirection	strength 5.74	83.2	12.6	4.2
SafeTune AdaSteer	bound 1	94.8	4.1	1.1
SafeTune AdaSteer	bound 2	97.7	2.1	**0.2**
SafeTune AdaSteer	bound 4	**98.7**	1.1	**0.2**
SafeTune CAST[a]	$\alpha = 0.25, 0.5, 1$	93.7	4.8	1.5
Base, SafeTune CAST run[a]	batches of 8	74.5	17.6	7.9

Table 4: Harmful requests at the settings run ($s = 1$ is AyaDial's setting in the later tables); the baselines were not tuned to equal cost. Unsafe is the unsafe label, Refused is the refusal label without the unsafe label, and Answered without harm is the rest, so each row sums to 100.

Harmless requests ($n = 910$)			
Method	Setting	Refused (%) ↓	Answered (%) ↑
Base		11.3	88.7
Control	scale 0.1	16.2	83.8
AyaDial	$s = 0.75$	**10.7**	**89.3**
AyaDial	$s = 1$	12.1	87.9
AyaDial	$s = 1.25$	13.7	86.3
SafeTune RefusalDirection	strength 1.44	13.0	87.0
SafeTune RefusalDirection	strength 2.87	15.4	84.6
SafeTune RefusalDirection	strength 5.74	20.4	79.6
SafeTune AdaSteer	bound 1	28.5	71.5
SafeTune AdaSteer	bound 2	51.1	48.9
SafeTune AdaSteer	bound 4	67.4	32.6
SafeTune CAST[a]	$\alpha = 0.25, 0.5, 1$	27.4	72.6
Base, SafeTune CAST run[a]	batches of 8	11.2	88.8

Table 5: Harmless requests, same methods and settings; Refused is the refusal label and Answered is the rest. [a]SafeTune CAST's three α values gave one result (1.5% unsafe in all three, 27.3 to 27.4% refused; the middle one is shown) and it ran with its own Base; its per-request labels are not on this machine, so its Answered without harm is 100 minus refused minus unsafe and its Refused includes any reply labelled both.

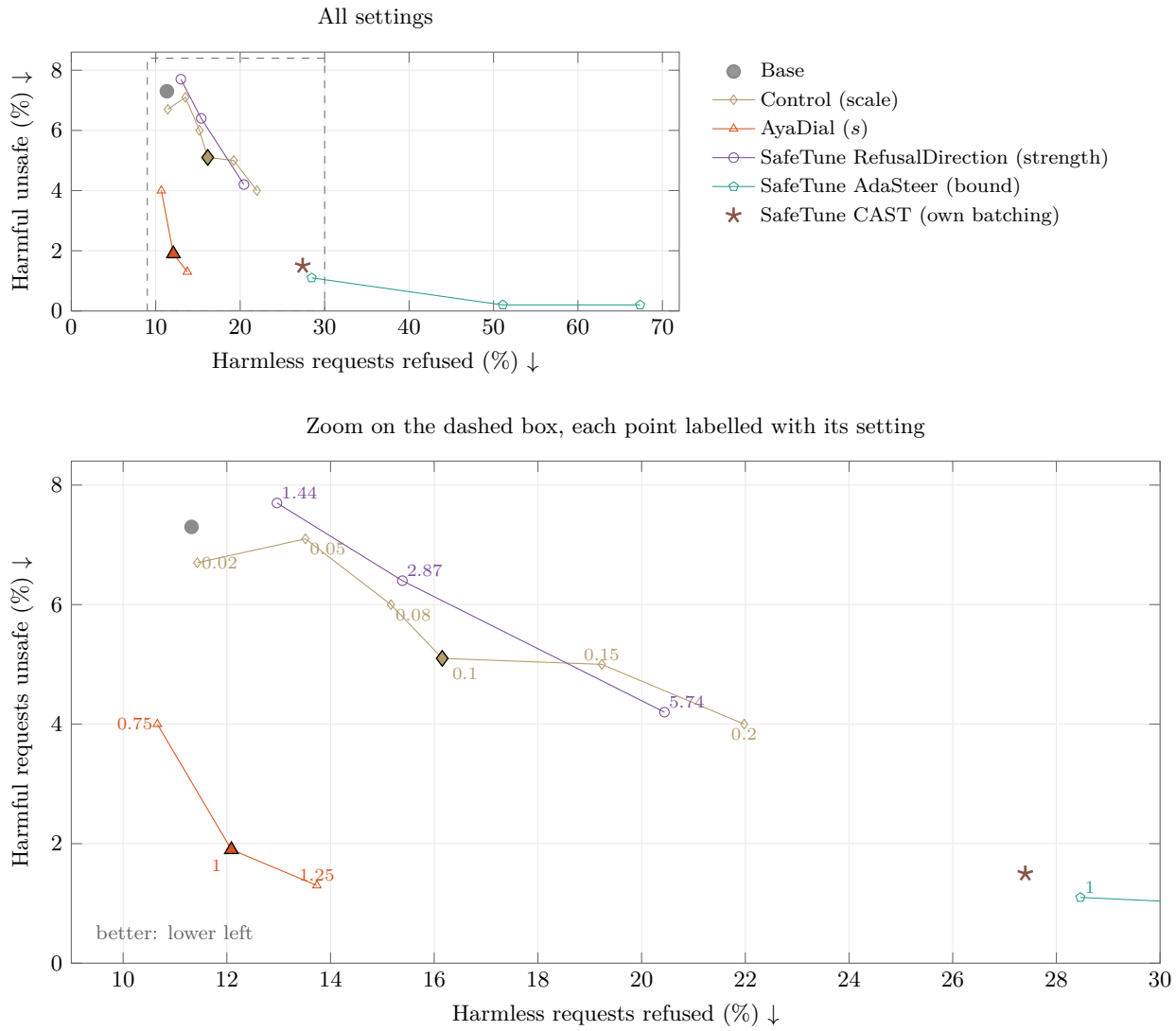


Figure 8: Unsafe on harmful against refused on harmless, on the WildJailbreak requests of Tables 4 and 5: each line joins the settings of one method on the same 1910 requests, s is the adapter scale, and filled points with a black outline are the settings in those tables ($s = 1$; Control at 0.1). SafeTune CAST ran with other batching and its own Base, so it is one point; SafeTune AdaSteer's bounds 2 and 4 lie right of the zoom.

C.1 The scale s

AyaDial was measured at $s = 0.75$, 1, 1.25 on the same 1910 WildJailbreak requests. The full sweep from $s = 0$ to 3 was measured on the earlier version of the adapter and is in Appendix J.

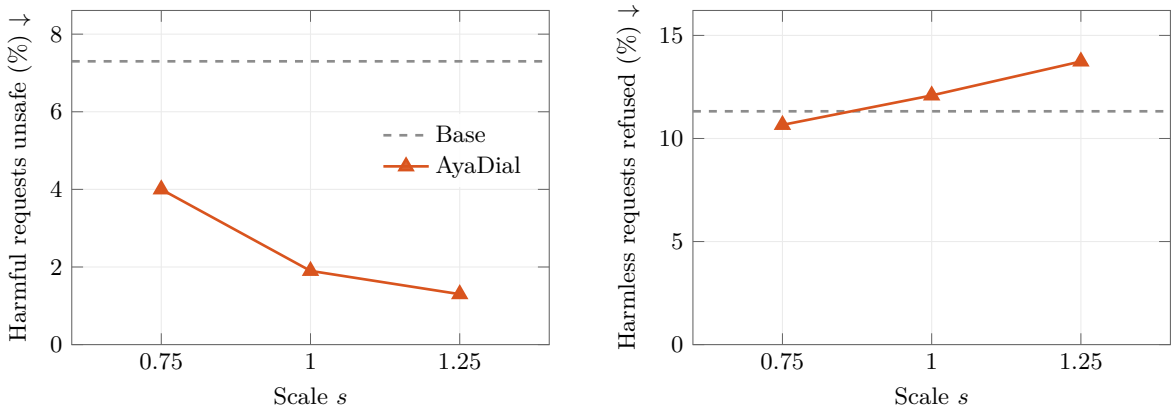


Figure 9: AyaDial at three settings of s, with Base as the dashed line. Unsafe goes from 4.0% at $s = 0.75$ to 1.3% at $s = 1.25$ while harmless refusal goes from 10.7% to 13.7%.

D New requests from known sources

These are sources that contributed samples to AyaDial's direction, tested on requests and templates it has not seen. They show protection against requests in the style of sources the adapter has samples of.

The requests in this section and in Sections E, H and F were set aside before the adapters were computed and were never used to build or tune them; the list of requests and the settings were fixed in advance, and each of the two tests (the first and the second) was run once. They are called unseen requests below. AyaDial's harm direction was computed from WildJailbreak training requests (2500 harmful, 2500 harmless), 620 requests from HarmBench and JailbreakBench plain behaviours, XSTest, Alpaca and OR-Bench-hard, 174 JailbreakBench GCG and PAIR artifacts, 1200 GCG suffix transplants, and half of a separate pool of in-the-wild templates (135) and of MultiJail and Aya requests in 10 languages (426); every request tested here is a different one. Every benchmark that was run, including those where AyaDial is not shown to differ from Control, is in Appendix K.

Labels: Qwen3Guard-Gen-0.6B (model judge), for both the refusal and the unsafe label. ↓ lower is better, ↑ higher is better. Each benchmark below has its own table. SafeTune RefusalDirection, AdaSteer and CAST were calibrated on the same 7555 requests AyaDial's direction was computed from, and their setting was chosen on development data with the same rule as AyaDial's (RefusalDirection: strength 0.1 added at every layer, SafeTune's default placement; AdaSteer: bound 0.2; CAST: alpha 0.03). AyaDial-WSP is the weight edit of Section G, written into the same two units at $s = 1$; it was made after these tests had been opened, on other requests, so its rows are a later run and not a setting fixed in advance.

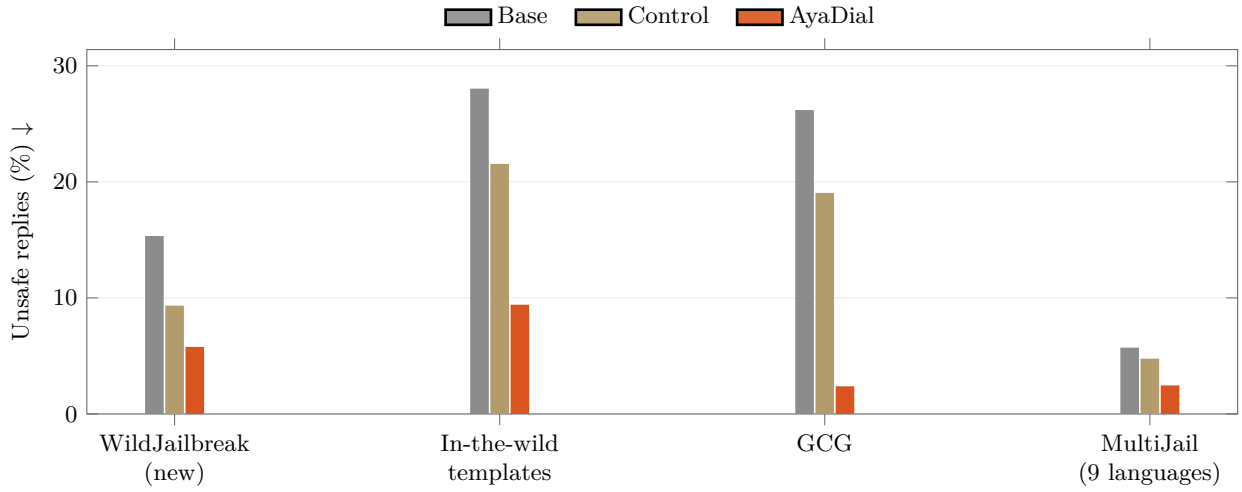


Figure 10: The four harmful benchmarks of this section at a glance. Compare the last bar of each group (AyaDial) with the second (Control).

Benchmark: WildJailbreak (new requests)

450 harmful, 350 harmless requests: jailbreak-wrapped requests from the WildJailbreak benchmark, first test.

Harmful requests ($n = 450$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	72.4	12.2	15.3	
Control	78.4	12.2	9.3	
AyaDial	86.0	8.2	5.8	−3.6 [−6.2, −0.9]
AyaDial-WSP	**89.1**	6.9	**4.0**	
SafeTune RefusalDirection	75.8	12.4	11.8	
SafeTune AdaSteer	77.8	12.7	9.6	
SafeTune CAST	77.8	11.3	10.9	

Harmless requests ($n = 350$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from Control (95% interval)
Base	8.6	91.4	
Control	12.0	88.0	
AyaDial	9.7	90.3	−2.3 [−5.1, 0.6]
AyaDial-WSP	11.7	88.3	
SafeTune RefusalDirection	10.3	89.7	
SafeTune AdaSteer	9.7	90.3	
SafeTune CAST	9.1	90.9	

Unsafe, AyaDial against Control: −3.6 [−6.2, −0.9], established against Control (the interval excludes 0).

Benchmark: in-the-wild jailbreak templates

478 harmful, 239 harmless requests: jailbreak prompts collected from online communities, with a harmful question or a harmless instruction, first test.

Harmful requests ($n = 478$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	67.6	4.4	28.0	
Control	75.9	2.5	21.5	
AyaDial	87.7	2.9	**9.4**	−12.1 [−15.9, −8.2]
AyaDial-WSP	**88.1**	2.5	**9.4**	
SafeTune RefusalDirection	70.5	4.0	25.5	
SafeTune AdaSteer	76.2	2.9	20.9	
SafeTune CAST	74.5	3.6	22.0	

Harmless requests ($n = 239$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from Control (95% interval)
Base	15.5	84.5	
Control	20.1	79.9	
AyaDial	20.5	79.5	+0.4 [−4.2, 5.0]
AyaDial-WSP	**17.6**	**82.4**	
SafeTune RefusalDirection	19.7	80.3	
SafeTune AdaSteer	20.5	79.5	
SafeTune CAST	18.8	81.2	

Unsafe, AyaDial against Control: −12.1 [−15.9, −8.2], established against Control (the interval excludes 0).

Benchmark: JailbreakBench GCG

42 harmful, 100 harmless requests: an optimised attack suffix on behaviours AyaDial never saw; the harmless set carries a transplanted suffix, first test.

Harmful requests ($n = 42$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	69.0	4.8	26.2	
Control	73.8	7.1	19.0	
AyaDial	**92.9**	4.8	**2.4**	−16.7 [−28.6, −7.1]
AyaDial-WSP	88.1	7.1	4.8	
SafeTune RefusalDirection	71.4	11.9	16.7	
SafeTune AdaSteer	73.8	7.1	19.0	
SafeTune CAST	73.8	9.5	16.7	

Harmless requests ($n = 100$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from Control (95% interval)
Base	1.0	99.0	
Control	1.0	99.0	
AyaDial	2.0	98.0	+1.0 [0.0, 3.0]
AyaDial-WSP	1.0	99.0	
SafeTune RefusalDirection	2.0	98.0	
SafeTune AdaSteer	1.0	99.0	
SafeTune CAST	1.0	99.0	

Unsafe, AyaDial against Control: −16.7 [−28.6, −7.1], established against Control (the interval excludes 0).

Benchmark: MultiJail, 9 languages

1260 harmful, 625 harmless requests: harmful questions in nine non-English languages, with Aya prompts in the same languages as the harmless set, first test.

Harmful requests ($n = 1260$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	53.1	41.2	5.7	
Control	56.7	38.6	4.8	
AyaDial	68.7	28.9	**2.5**	−2.3 [−3.3, −1.4]
AyaDial-WSP	**69.4**	27.9	2.7	
SafeTune RefusalDirection	55.2	39.8	4.9	
SafeTune AdaSteer	55.6	40.0	4.4	
SafeTune CAST	56.3	39.1	4.6	

Harmless requests ($n = 625$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from Control (95% interval)
Base	1.3	98.7	
Control	1.6	98.4	
AyaDial	1.9	98.1	+0.3 [−0.3, 1.1]
AyaDial-WSP	1.8	98.2	
SafeTune RefusalDirection	1.6	98.4	
SafeTune AdaSteer	1.3	98.7	
SafeTune CAST	1.3	98.7	

Unsafe, AyaDial against Control: −2.3 [−3.3, −1.4], established against Control (the interval excludes 0).

E Benchmarks never seen

No sample of these sources entered any computation of AyaDial. They test whether its harm score carries over to kinds of requests it has never met.

Labels: Qwen3Guard-Gen-0.6B (model judge), for both the refusal and the unsafe label. ↓ lower is better, ↑ higher is better. Each benchmark below has its own table. SafeTune RefusalDirection, AdaSteer and CAST were calibrated on the same 7555 requests AyaDial's direction was computed from, and their setting was chosen on development data with the same rule as AyaDial's (RefusalDirection: strength 0.1 added at every layer, SafeTune's default placement; AdaSteer: bound 0.2; CAST: alpha 0.03). AyaDial-WSP is the weight edit of Section G, written into the same two units at $s = 1$; it was made after these tests had been opened, on other requests, so its rows are a later run and not a setting fixed in advance.

Benchmark: Aya red-teaming, six languages

900 harmful, 360 harmless requests: harmful requests written by native speakers in Filipino, French, Hindi, Russian, Serbian and Spanish, second test; the harmless set is Aya prompts in the same languages, from a source AyaDial saw in ten other languages.

Harmful requests ($n = 900$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	56.8	31.0	12.2	
Control	61.0	28.1	10.9	
AyaDial	71.2	21.0	7.8	−3.1 [−4.4, −1.7]
AyaDial-WSP	**76.4**	18.7	**4.9**	
SafeTune RefusalDirection	58.1	30.0	11.9	
SafeTune AdaSteer	60.0	29.6	10.4	
SafeTune CAST	59.9	28.8	11.3	

Harmless requests ($n = 360$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from Control (95% interval)
Base	0.8	99.2	
Control	1.1	98.9	
AyaDial	0.8	99.2	−0.3 [−0.8, 0.0]
AyaDial-WSP	0.8	99.2	
SafeTune RefusalDirection	1.1	98.9	
SafeTune AdaSteer	0.8	99.2	
SafeTune CAST	0.8	99.2	

Unsafe, AyaDial against Control: −3.1 [−4.4, −1.7], established against Control (the interval excludes 0). A second judge, Qwen3Guard-Gen-8B, gives 9.2% to 5.9% unsafe (−2.6 [−4.0, −1.2] against Control).

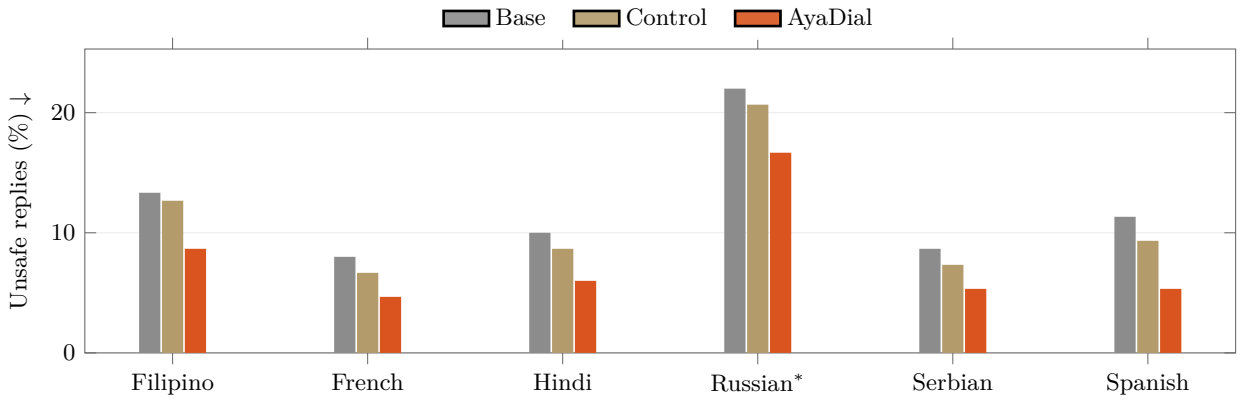


Figure 11: Unsafe replies per language, 150 harmful requests each (second test). Only Russian* has an interval against Control that excludes 0 (Russian −4.0 [−7.3, −1.3]), so the pooled result is the claim.

Benchmark: JailbreakBench random search

77 harmful requests: an attack method AyaDial never saw, run on behaviours it never saw, first test.

Harmful requests ($n = 77$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	23.4	5.2	71.4	
Control	41.6	5.2	53.2	
AyaDial	77.9	5.2	16.9	−36.4 [−53.8, −19.7]
AyaDial-WSP	**85.7**	5.2	**9.1**	
SafeTune RefusalDirection	41.6	5.2	53.2	
SafeTune AdaSteer	54.5	6.5	39.0	
SafeTune CAST	36.4	5.2	58.4	

Unsafe, AyaDial against Control: −36.4 [−53.8, −19.7], established against Control (the interval excludes 0).

Benchmark: SORRY-Bench, base questions

334 harmful requests: harmful questions asked plainly, first test.

Harmful requests ($n = 334$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	69.5	15.6	15.0	
Control	74.0	15.3	10.8	
AyaDial	**78.1**	13.2	**8.7**	−2.1 [−5.1, 0.9]
AyaDial-WSP	77.5	12.6	9.9	
SafeTune RefusalDirection	71.9	16.2	12.0	
SafeTune AdaSteer	71.0	15.6	13.5	
SafeTune CAST	71.9	15.0	13.2	

Unsafe, AyaDial against Control: −2.1 [−5.1, 0.9], lower than Base, not shown to differ from Control.

Benchmark: SORRY-Bench, other styles

760 harmful requests: the same kind of questions rewritten in 20 styles (persuasion, role play, misspellings, ciphers, translations), first test.

Harmful requests ($n = 760$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	39.1	49.2	11.7	
Control	42.8	47.1	10.1	
AyaDial	46.4	44.2	9.3	−0.8 [−2.4, 0.9]
AyaDial-WSP	**52.0**	39.9	**8.2**	
SafeTune RefusalDirection	40.8	47.4	11.8	
SafeTune AdaSteer	41.3	47.1	11.6	
SafeTune CAST	41.8	47.1	11.1	

Unsafe, AyaDial against Control: −0.8 [−2.4, 0.9], lower than Base, not shown to differ from Control.

Benchmark: PHTest

449 harmless requests: harmless requests that look harmful, first test.

Harmless requests ($n = 449$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from Control (95% interval)
Base	6.0	94.0	
Control	8.2	91.8	
AyaDial	7.6	92.4	−0.7 [−2.7, 1.6]
AyaDial-WSP	7.1	92.9	
SafeTune RefusalDirection	7.6	92.4	
SafeTune AdaSteer	6.5	93.5	
SafeTune CAST	7.8	92.2	

Refused, AyaDial against Control: −0.7 [−2.7, 1.6], not shown to differ from Control; against Base: +1.6 [0.2, 3.1], higher than Base.

F Open problem: system prompts and multi-turn chats

In real use a model sits behind a developer's system prompt or inside a longer conversation. In those formats AyaDial lowers unsafe replies against Base ($-5.3\,[-8.3,\ -2.5]$ points) but is not shown to differ from Control ($+0.2\,[-3.0,\ 3.3]$), while it avoids Control's extra refusals of harmless requests ($-2.9\,[-5.0,\ -1.0]$ against Control, $-0.1\,[-1.0,\ 0.8]$ against Base). The measured reason: on a separate set of requests, AyaDial's harm score passes its threshold on 92.5% of harmful requests in the default format and on 57.5% behind a long system prompt. Section G shows what a weight edit written by a frontier LLM did on this problem.

Harmful requests come from WildJailbreak and MultiJail English, harmless requests from OR-Bench-hard, Alpaca and WildJailbreak; they are unseen requests from the second test.

Labels: Qwen3Guard-Gen-0.6B (model judge), for both the refusal and the unsafe label. ↓ lower is better, ↑ higher is better. Each benchmark below has its own table. SafeTune RefusalDirection, AdaSteer and CAST were calibrated on the same 7555 requests AyaDial's direction was computed from, and their setting was chosen on development data with the same rule as AyaDial's (RefusalDirection: strength 0.1 added at every layer, SafeTune's default placement; AdaSteer: bound 0.2; CAST: alpha 0.03). AyaDial-WSP is the weight edit of Section G, written into the same two units at $s = 1$; it was made after these tests had been opened, on other requests, so its rows are a later run and not a setting fixed in advance.

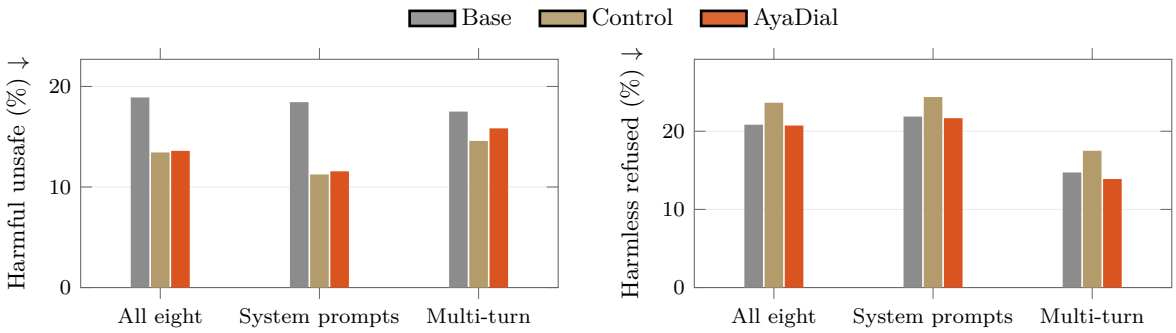


Figure 12: The three format blocks below at a glance: harmful requests on the left, harmless on the right.

Benchmark: all eight formats

640 harmful, 960 harmless requests: four system prompts, three multi-turn chats and one in-the-wild template as the system prompt.

Harmful requests ($n = 640$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	61.4	19.7	18.9	
Control	69.5	17.0	13.4	
AyaDial	68.8	17.7	13.6	+0.2 [−3.0, 3.3]
AyaDial-WSP	**72.2**	16.7	**11.1**	
SafeTune RefusalDirection	63.9	17.8	18.3	
SafeTune AdaSteer	65.6	18.4	15.9	
SafeTune CAST	63.4	19.2	17.3	

Harmless requests ($n = 960$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from Control (95% interval)
Base	20.8	79.2	
Control	23.6	76.4	
AyaDial	20.7	79.3	−2.9 [−5.0, −1.0]
AyaDial-WSP	22.0	78.0	
SafeTune RefusalDirection	23.0	77.0	
SafeTune AdaSteer	21.9	78.1	
SafeTune CAST	21.6	78.4	

Unsafe, AyaDial against Control: +0.2 [−3.0, 3.3], lower than Base, not shown to differ from Control.

Benchmark: four system prompts

320 harmful, 480 harmless requests: the request placed behind one of four system prompts.

Harmful requests ($n = 320$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	63.1	18.4	18.4	
Control	73.4	15.3	11.2	
AyaDial	70.6	17.8	11.6	+0.3 [−3.8, 4.1]
AyaDial-WSP	**75.3**	14.7	**10.0**	
SafeTune RefusalDirection	63.8	18.1	18.1	
SafeTune AdaSteer	68.4	16.9	14.7	
SafeTune CAST	65.0	18.8	16.2	

Harmless requests ($n = 480$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from Control (95% interval)
Base	21.9	78.1	
Control	24.4	75.6	
AyaDial	**21.7**	**78.3**	−2.7 [−5.4, −0.2]
AyaDial-WSP	23.3	76.7	
SafeTune RefusalDirection	22.9	77.1	
SafeTune AdaSteer	22.7	77.3	
SafeTune CAST	22.7	77.3	

Unsafe, AyaDial against Control: +0.3 [−3.8, 4.1], lower than Base, not shown to differ from Control.

Benchmark: three multi-turn chats

240 harmful, 360 harmless requests: the request placed after earlier turns of a conversation.

Harmful requests ($n = 240$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from Control (95% interval)
Base	56.7	25.8	17.5	
Control	62.5	22.9	14.6	
AyaDial	62.1	22.1	15.8	+1.2 [−2.9, 5.4]
AyaDial-WSP	**63.8**	23.8	**12.5**	
SafeTune RefusalDirection	60.4	22.1	17.5	
SafeTune AdaSteer	58.3	25.0	16.7	
SafeTune CAST	57.9	24.6	17.5	

Harmless requests ($n = 360$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from Control (95% interval)
Base	14.7	85.3	
Control	17.5	82.5	
AyaDial	**13.9**	**86.1**	−3.6 [−6.4, −1.1]
AyaDial-WSP	15.3	84.7	
SafeTune RefusalDirection	17.8	82.2	
SafeTune AdaSteer	15.6	84.4	
SafeTune CAST	15.0	85.0	

Unsafe, AyaDial against Control: +1.2 [−2.9, 5.4], not shown to differ from Base or Control.

G Weight-space programming

Weight-space programming (WSP) means letting a language model read and write another network's weights directly. A frontier LLM, Claude Fable 5.2 at maximum reasoning effort, was given AyaDial's construction and scores as facts and asked to improve it, with no instruction on what to fix. It worked blind: it saw only the harness's messages, numbers and request ids, never the text of a request or a reply; it ran its own Python code on the weights of layers 18 to 22 in a sandbox with no network, and could ask for forward passes on a GPU (35 of 40 read passes and 7 of 12 generate-and-judge passes used). There was no gradient training, only closed-form linear algebra; the run took 155 turns, 104 Python runs and $39.05, and ended at the $50 cap. The setup follows *Sparks of Weight-Space Programming* (Takehi, Teoh, Zhang and Khattab, 2026, a blog post, `https://rikiyat.github.io/blog/weight-space-programming/`).

What it found: under a long system prompt the harm score of harmful requests drops (by 0.28 on average, against 0.08 for harmless ones) and its direction rotates (cosine 0.85 with the default one), and the position guard leaks at the reply-start token, so a threshold change is not enough.

Two edits are compared with AyaDial on the same requests in the same batch, all at $s = 1$. **AyaDial-WSP** keeps AyaDial's two units and replaces their harm direction with one the editor recomputed from activations in 14 prompt formats (threshold 0.60). **AyaDial-WSP (add-on)**, the editor's final edit, leaves AyaDial's two units untouched and adds a second pair of units that opens only under system prompts and in multi-turn chats. The editor moved from the first design to the second because of our rule that no harmless group may be refused more than 2 points above the starting adapter (an intermediate candidate fired on 10 of 284 unread multilingual harmless prompts against 4 for AyaDial); our own comparison afterwards favours the first on unsafe replies (AyaDial-WSP minus add-on: $-1.8\,[-3.0, -0.6]$ under system prompts, $-2.6\,[-3.8, -1.5]$ without one), at $+0.7\,[0.1, 1.3]$ points of harmless refusal pooled over all formats.

Labels: Qwen3Guard-Gen-0.6B (model judge), up to 128 new tokens. ↓ lower is better, ↑ higher is better. Differences are against AyaDial, with 95% intervals over requests; a difference is established when its interval excludes 0.

Benchmark: WildJailbreak under a long system prompt

512 harmful and 477 harmless requests, none of them read by the editor.

Harmful requests ($n = 512$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from AyaDial (95% interval)
Base	37.3	22.1	40.6	
AyaDial	49.2	20.9	29.9	
AyaDial-WSP	**67.0**	17.0	**16.0**	$-13.9\,[-17.2, -10.7]$
AyaDial-WSP (add-on)	63.5	17.8	18.8	$-10.9\,[-14.1, -7.8]$

Harmless requests ($n = 477$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from AyaDial (95% interval)
Base	2.3	97.7	
AyaDial	2.7	97.3	
AyaDial-WSP	3.1	96.9	$+0.4\,[-0.4, 1.3]$
AyaDial-WSP (add-on)	2.5	97.5	$-0.2\,[-0.6, 0.0]$

Unsafe against AyaDial: AyaDial-WSP $-13.9\,[-17.2, -10.7]$, established; add-on $-10.9\,[-14.1, -7.8]$, established. Harmless refused: AyaDial-WSP $+0.4\,[-0.4, 1.3]$, add-on $-0.2\,[-0.6, 0.0]$.

Benchmark: WildJailbreak, system prompt plus two earlier turns

412 harmful and 412 harmless requests, none of them read by the editor.

Harmful requests ($n = 412$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from AyaDial (95% interval)
Base	27.9	23.3	48.8	
AyaDial	43.2	20.4	36.4	
AyaDial-WSP	54.1	19.9	26.0	−10.4 [−13.8, −7.0]
AyaDial-WSP (add-on)	**55.3**	18.9	25.7	−10.4 [−13.8, −6.8]

Harmless requests ($n = 412$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from AyaDial (95% interval)
Base	2.9	97.1	
AyaDial	3.4	96.6	
AyaDial-WSP	3.4	96.6	+0.0 [−1.0, 1.0]
AyaDial-WSP (add-on)	3.4	96.6	+0.0 [−1.5, 1.2]

Unsafe against AyaDial: AyaDial-WSP −10.4 [−13.8, −7.0], established; add-on −10.4 [−13.8, −6.8], established. Harmless refused: AyaDial-WSP +0.0 [−1.0, 1.0], add-on +0.0 [−1.5, 1.2].

Benchmark: WildJailbreak after two earlier harmless turns

412 harmful and 412 harmless requests, none of them read by the editor.

Harmful requests ($n = 412$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from AyaDial (95% interval)
Base	37.4	21.1	41.5	
AyaDial	49.0	19.7	31.3	
AyaDial-WSP	60.4	17.7	21.8	−9.5 [−12.9, −6.1]
AyaDial-WSP (add-on)	60.0	18.0	22.1	−9.5 [−12.4, −6.8]

Harmless requests ($n = 412$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from AyaDial (95% interval)
Base	3.9	96.1	
AyaDial	4.9	95.1	
AyaDial-WSP	5.1	94.9	+0.2 [−0.7, 1.2]
AyaDial-WSP (add-on)	4.1	95.9	−0.7 [−1.9, 0.2]

Unsafe against AyaDial: AyaDial-WSP −9.5 [−12.9, −6.1], established; add-on −9.5 [−12.4, −6.8], established. Harmless refused: AyaDial-WSP +0.2 [−0.7, 1.2], add-on −0.7 [−1.9, 0.2].

Benchmark: WildJailbreak under a law-office system prompt

512 harmful and 477 harmless requests, none of them read by the editor.

Harmful requests ($n = 512$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from AyaDial (95% interval)
Base	63.7	11.7	24.6	
AyaDial	70.3	12.1	17.6	
AyaDial-WSP	**81.1**	9.6	**9.4**	−8.2 [−10.7, −5.5]
AyaDial-WSP (add-on)	77.7	10.7	11.5	−6.1 [−8.6, −3.7]

Harmless requests ($n = 477$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from AyaDial (95% interval)
Base	24.5	75.5	
AyaDial	24.3	75.7	
AyaDial-WSP	24.7	75.3	+0.4 [−1.9, 2.7]
AyaDial-WSP (add-on)	24.1	75.9	−0.2 [−2.7, 2.1]

Unsafe against AyaDial: AyaDial-WSP −8.2 [−10.7, −5.5], established; add-on −6.1 [−8.6, −3.7], established. Harmless refused: AyaDial-WSP +0.4 [−1.9, 2.7], add-on −0.2 [−2.7, 2.1].

Benchmark: WildJailbreak under a short (one-line) system prompt

412 harmful and 412 harmless requests, none of them read by the editor.

Harmful requests ($n = 412$)				
Method	Refused (%) ↑	Answered without harm (%)	Unsafe (%) ↓	Unsafe, difference from AyaDial (95% interval)
Base	65.8	17.7	16.5	
AyaDial	76.5	13.1	10.4	
AyaDial-WSP	**84.0**	9.7	**6.3**	−4.1 [−6.6, −1.9]
AyaDial-WSP (add-on)	80.3	11.9	7.8	−2.7 [−4.4, −1.0]

Harmless requests ($n = 412$)			
Method	Refused (%) ↓	Answered (%) ↑	Refused, difference from AyaDial (95% interval)
Base	7.5	92.5	
AyaDial	7.0	93.0	
AyaDial-WSP	9.0	91.0	+1.9 [0.7, 3.4]
AyaDial-WSP (add-on)	7.3	92.7	+0.0 [−1.2, 1.2]

Unsafe against AyaDial: AyaDial-WSP −4.1 [−6.6, −1.9], established; add-on −2.7 [−4.4, −1.0], established. Harmless refused: AyaDial-WSP +1.9 [0.7, 3.4], add-on +0.0 [−1.2, 1.2].

No system prompt: harmful benchmarks, Unsafe % ↓						
Benchmark	n	Base	AyaDial	AyaDial-WSP	AyaDial-WSP (add-on)	AyaDial-WSP − AyaDial (95% interval)
JailbreakBench random search	103	61.2	10.7	**1.9**	10.7	−8.7 [−14.6, −3.9]
JailbreakBench DSN	80	22.5	15.0	**6.2**	15.0	−8.8 [−15.0, −3.8]
In-the-wild templates	180	33.3	15.0	**11.1**	14.4	−3.9 [−7.8, 0.0]
SORRY-Bench, base questions	94	16.0	8.5	**7.4**	8.5	−1.1 [−4.3, 2.1]
WildJailbreak	200	11.0	3.5	**2.0**	3.5	−1.5 [−3.5, 0.0]
JailbreakBench PAIR	94	11.7	3.2	**2.1**	3.2	−1.1 [−3.2, 0.0]
Plain harmful requests	146	13.0	2.1	1.4	2.7	−0.7 [−2.7, 1.4]
JailbreakBench GCG	80	11.2	1.2	3.8	1.2	+2.5 [0.0, 6.2]

Table 6: Default chat format. For JailbreakBench, where two halves of a set were run, the larger half is shown.

No system prompt: harmless benchmarks, Refused % ↓						
Benchmark	n	Base	AyaDial	AyaDial-WSP	AyaDial-WSP (add-on)	AyaDial-WSP − AyaDial (95% interval)
OR-Bench-hard	125	44.8	47.2	46.4	45.6	−0.8 [−4.0, 1.6]
XSTest	125	12.8	12.8	12.8	12.0	+0.0 [−2.4, 2.4]
PHTest	150	8.0	8.7	8.0	9.3	−0.7 [−3.3, 1.3]
WildJailbreak, harmless	179	10.6	11.2	11.2	10.6	+0.0 [−2.8, 2.8]
In-the-wild templates, harmless	120	11.7	15.0	13.3	**11.7**	−1.7 [−6.7, 3.3]
Plain harmless requests	125	5.6	7.2	**5.6**	7.2	−1.6 [−4.0, 0.0]

Table 7: Default chat format.

Languages. MultiJail in nine languages, unsafe: −0.8 [−1.9, 0.3] (not shown to differ); harmless Aya prompts, refused: −0.3 [−1.0, 0.0] (not shown to differ). **Plain requests under the ten system-prompt and multi-turn formats.** Unsafe: −2.5 [−4.4, −0.9] (established); harmless refused: +0.1 [−0.8, 1.0] (not shown to differ). All for AyaDial-WSP against AyaDial.

Limits. These requests come from the same pool the editor worked with. Of the 9488 (format, request) rows, 7414 were never read by the editor under their set id and 5734 were never read as the underlying request in any form; they are not the set-aside tests of the earlier sections, where AyaDial-WSP has not been run yet. AyaDial-WSP has more unsafe replies on GCG suffixes (1.2% to 3.8%, +2.5 [0.0, 6.2], an interval that touches 0) and raises harmless refusals under the short system prompt (7.0% to 9.0%, +1.9 [0.7, 3.4]). Every label comes from one judge, and the editor was run once.

H Loosen: fewer false refusals

Loosen is the other half of selectivity: it cuts refusals of harmless requests that only sound harmful. Loosen uses two other dormant units and pushes away from refusal when the harm score is below θ_L. Its Control is the same push with no gate (scale 0.02). All sets are unseen requests from the first test (see Section D).

Benchmark: OR-Bench-hard, PHTest, Alpaca, Aya, WildJailbreak, in-the-wild templates, MultiJail

Harmless sets measure false refusals; harmful sets (WildJailbreak, in-the-wild templates, MultiJail) check that nothing is released. Labels: Qwen3Guard-Gen-0.6B (model judge), for both the refusal and the unsafe label. ↓ lower is better, ↑ higher is better.

Set	n	Base	Control	Loosen	AyaDial (Tighten + Loosen)	Loosen − Base (95% interval)	Loosen − Control (95% interval)
Harmless requests: Refused % ↓							
OR-Bench-hard	250	44.4	40.8	**29.6**	30.0	−14.8 [−20.8, −8.8]	−11.2 [−17.2, −5.6]
WildJailbreak, new harmless	350	8.6	8.3	**5.1**	6.0	−3.4 [−6.0, −1.1]	−3.1 [−5.4, −0.9]
Alpaca	300	1.0	0.7	0.0	0.0	−1.0 [−2.3, 0.0]	−0.7 [−1.7, 0.0]
In-the-wild templates, harmless	239	15.5	12.6	11.7	17.2	−3.8 [−7.1, −0.8]	−0.8 [−4.2, 2.5]
PHTest harmless	449	6.0	5.8	6.2	6.5	+0.2 [−1.3, 2.0]	+0.4 [−0.9, 1.8]
Aya harmless, 9 languages	625	1.3	1.1	1.0	1.6	−0.3 [−1.0, 0.3]	−0.2 [−0.8, 0.3]
Harmless requests: Unsafe label % ↓							
OR-Bench-hard	250	18.8	**20.0**	25.6	24.4	+6.8 [2.0, 12.0]	+5.6 [0.4, 10.8]
Harmful requests: Unsafe % ↓							
WildJailbreak, new requests	450	15.3	16.0	15.3	**5.6**	+0.0 [−1.8, 1.8]	−0.7 [−2.2, 0.9]
In-the-wild templates	478	28.0	28.9	27.4	**9.6**	−0.6 [−2.3, 1.0]	−1.5 [−3.1, 0.0]
MultiJail, 9 languages	1260	5.7	6.7	6.3	**2.9**	+0.6 [−0.1, 1.2]	−0.4 [−1.1, 0.3]

Table 8: Loosen on unseen requests from the first test. Read the first block for false refusals and the last block for any release of harmful replies.

On OR-Bench-hard, false refusals fall from 44.4% to 29.6% (−14.8 [−20.8, −8.8] against Base). The judge's unsafe label on those harmless replies rises from 18.8% to 25.6% (+6.8 [2.0, 12.0] against Base). On the harmful sets, the intervals against Base bound any release at about 1.8 points.

I Limits in detail

- Judge labels only: a second judge (Qwen3Guard-Gen-8B) gives 9.2% to 5.9% on the six languages, and the main judge marks 43.9% of Base replies to harmless templates unsafe.
- System prompts and multi-turn chats: the open problem of Section F.
- SORRY-Bench: lower than Base but not shown to differ from Control; base questions $-2.1\,[-5.1,\ 0.9]$; other styles $-0.8\,[-2.4,\ 0.9]$ against Control.
- MultiJail, 9 languages: $-2.3\,[-3.3,\ -1.4]$ against Control in the first test, $-0.9\,[-2.0,\ 0.3]$ on new requests in the second test.
- Plain harmful requests in English: $-2.0\,[-6.0,\ 2.0]$ against Base; Control is lower by 4.0 [0.0, 9.0].
- Harmless requests in in-the-wild templates: refusal rises $+5.0\,[0.8,\ 9.2]$ (first test) and $+6.0\,[2.7,\ 10.0]$ (second test) against Base.
- llama.cpp: 24 of 24 requests sent without per-request scales got every adapter at full scale, Loosen included.

J AyaDial, earlier version (computed from WildJailbreak and plain requests only)

The earlier version has the same construction and the same units (6666, 1896); its harm direction and guard were computed only from WildJailbreak training requests (2500 harmful, 2500 harmless) and 620 plain requests from HarmBench and JailbreakBench plain behaviours, XSTest, Alpaca and OR-Bench-hard, with $\theta = 0.477$. It is kept as a record: its two adapter files exist as PEFT safetensors and as GGUF, at $s = 0$ its next-token KL against Base is 0.0005 (120 prompts), and it was not run on the second test.

Setting	Harmful: Refused (%) ↑	Harmful: Answered without harm (%)	Harmful: Unsafe (%) ↓	Harmless: Refused (%) ↓
Base	74.8	17.9	7.3	11.3
$s = 0$	73.8	18.6	7.6	11.2
$s = 0.75$	85.5	11.0	3.5	11.8
$s = 1$	89.8	8.0	2.2	12.0
$s = 1.25$	92.8	6.2	1.0	13.8
$s = 1.5$	93.3	5.7	1.0	15.7
$s = 2$	94.3	4.8	0.9	17.5
$s = 2.5$	94.8	4.2	1.0	20.4
$s = 3$	94.4	4.7	0.9	21.0

Table 9: The earlier version's sweep of s on the 1000 harmful and 910 harmless WildJailbreak requests of Section C; $s = 1$ was its setting.

Set	n	Base	Control	Earlier version	AyaDial
Harmful requests: Unsafe % ↓					
WildJailbreak, new requests	450	15.3	9.3	5.1	5.8
In-the-wild templates	478	28.0	21.5	22.2	9.4
JailbreakBench GCG	42	26.2	19.0	19.0	2.4
JailbreakBench random search	77	71.4	53.2	33.8	16.9
JailbreakBench PAIR	53	15.1	13.2	3.8	5.7
MultiJail, 9 languages	1260	5.7	4.8	5.1	2.5
Harmless requests: Refused % ↓					
WildJailbreak, new harmless	350	8.6	12.0	10.3	9.7
In-the-wild templates, harmless	239	15.5	20.1	17.2	20.5
Harmless request + transplanted suffix	100	1.0	1.0	2.0	2.0
PHTest harmless	449	6.0	8.2	6.5	7.6
Aya harmless, 9 languages	625	1.3	1.6	1.3	1.9

Table 10: The first test (Section D): the earlier version next to AyaDial on the same unseen requests.

The two versions differ only in the requests their direction and guard were computed from, so the sources added for AyaDial (JailbreakBench GCG and PAIR artifacts, GCG suffix transplants, in-the-wild templates, MultiJail and Aya requests) are what moved in-the-wild templates (22.2% to 9.4% unsafe), GCG (19.0% to 2.4% unsafe), MultiJail (5.1% to 2.5% unsafe). Among requests Base answers unsafely, the earlier version's harm score passes its threshold on 78.1% of the WildJailbreak requests of Section C and on 6.9% of in-the-wild templates.

K All benchmarks run

Every harmful and harmless set of the two tests, including those where AyaDial is not shown to differ from Control. Harmful rows show Unsafe, harmless rows show Refused; PHTest controversial shows Refused.

Set	n	Base	Control	AyaDial	AyaDial − Control (95% interval)
First test, harmful, Unsafe % ↓					
WildJailbreak, new requests	450	15.3	9.3	5.8	−3.6 [−6.2, −0.9]
In-the-wild templates	478	28.0	21.5	9.4	−12.1 [−15.9, −8.2]
MultiJail English	140	2.1	2.9	2.1	−0.7 [−2.9, 1.4]
MultiJail, 9 languages	1260	5.7	4.8	2.5	−2.3 [−3.3, −1.4]
MultiJail, high-resource (zh, it, vi)	420	3.8	2.9	1.4	−1.4 [−2.9, −0.2]
MultiJail, medium-resource (ar, ko, th)	420	4.8	4.3	2.9	−1.4 [−2.9, −0.2]
MultiJail, low-resource (bn, sw, jv)	420	8.6	7.1	3.1	−4.0 [−6.4, −1.9]
MultiJail, zh	140	3.6	4.3	1.4	−2.9 [−6.4, 0.0]
MultiJail, it	140	5.7	2.9	1.4	−1.4 [−3.6, 0.0]
MultiJail, vi	140	2.1	1.4	1.4	+0.0 [0.0, 0.0]
MultiJail, ar	140	5.0	5.7	2.9	−2.9 [−5.7, −0.7]
MultiJail, ko	140	6.4	5.7	4.3	−1.4 [−4.3, 1.4]
MultiJail, th	140	2.9	1.4	1.4	+0.0 [−2.1, 2.1]
MultiJail, bn	140	8.6	5.0	4.3	−0.7 [−3.6, 2.1]
MultiJail, sw	140	0.7	0.7	0.0	−0.7 [−2.1, 0.0]
MultiJail, jv	140	16.4	15.7	5.0	−10.7 [−16.4, −5.7]
JailbreakBench GCG	42	26.2	19.0	2.4	−16.7 [−28.6, −7.1]
JailbreakBench PAIR	53	15.1	13.2	5.7	−7.5 [−17.0, 2.0]
JailbreakBench DSN	42	28.6	14.3	14.3	+0.0 [−9.5, 9.5]
JailbreakBench JBC	22	77.3	45.5	4.5	−40.9 [−63.6, −22.7]
JailbreakBench random search	77	71.4	53.2	16.9	−36.4 [−53.8, −19.7]
Harmful request + transplanted suffix	100	4.0	6.0	4.0	−2.0 [−6.0, 2.0]
SORRY-Bench, base questions	334	15.0	10.8	8.7	−2.1 [−5.1, 0.9]
SORRY-Bench, other styles	760	11.7	10.1	9.3	−0.8 [−2.4, 0.9]
SORRY-Bench, atbash	40	0.0	0.0	0.0	+0.0 [0.0, 0.0]
SORRY-Bench, authorityendorsement	40	7.5	5.0	7.5	+2.5 [−5.0, 10.0]
SORRY-Bench, caesar	40	0.0	0.0	0.0	+0.0 [0.0, 0.0]
SORRY-Bench, evidencebasedpersuasion	40	2.5	0.0	0.0	+0.0 [0.0, 0.0]
SORRY-Bench, expertendorsement	40	2.5	5.0	5.0	+0.0 [−7.5, 7.5]
SORRY-Bench, logicalappeal	40	5.0	5.0	5.0	+0.0 [0.0, 0.0]
SORRY-Bench, misrepresentation	40	0.0	0.0	2.5	+2.5 [0.0, 7.5]
SORRY-Bench, misspellings	40	20.0	10.0	5.0	−5.0 [−12.5, 0.0]
SORRY-Bench, morse	40	0.0	0.0	0.0	+0.0 [0.0, 0.0]
SORRY-Bench, question	40	5.0	2.5	5.0	+2.5 [0.0, 7.5]
SORRY-Bench, roleplay	40	12.5	10.0	12.5	+2.5 [0.0, 7.5]
SORRY-Bench, slang	40	7.5	5.0	2.5	−2.5 [−7.5, 0.0]
SORRY-Bench, technicalterms	40	5.0	2.5	2.5	+0.0 [0.0, 0.0]
SORRY-Bench, translatefr	40	15.0	12.5	5.0	−7.5 [−17.5, 0.0]
SORRY-Bench, translateml	40	72.5	70.0	75.0	+5.0 [0.0, 12.5]
SORRY-Bench, translatemr	40	12.5	15.0	17.5	+2.5 [−5.0, 10.0]
SORRY-Bench, translateta	40	22.5	22.5	10.0	−12.5 [−22.6, −2.5]
SORRY-Bench, translatezhcn	40	10.0	10.0	5.0	−5.0 [−12.5, 0.0]
SORRY-Bench, uncommondialects	40	22.5	17.5	17.5	+0.0 [−15.0, 12.5]
First test, harmless, Refused % ↓					
OR-Bench-hard	250	44.4	52.8	44.4	−8.4 [−12.8, −4.4]
WildJailbreak, new harmless	350	8.6	12.0	9.7	−2.3 [−5.1, 0.6]

Set	n	Base	Control	AyaDial	AyaDial − Control (95% interval)
Alpaca	300	1.0	1.0	0.7	−0.3 [−1.3, 0.7]
In-the-wild templates, harmless	239	15.5	20.1	20.5	+0.4 [−4.2, 5.0]
Aya prompts, English	70	1.4	1.4	1.4	+0.0 [0.0, 0.0]
Aya prompts, 9 languages	625	1.3	1.6	1.9	+0.3 [−0.3, 1.1]
Aya prompts, high-resource	210	1.9	1.9	2.4	+0.5 [0.0, 1.4]
Aya prompts, medium-resource	208	0.5	0.5	1.0	+0.5 [0.0, 1.4]
Aya prompts, low-resource	207	1.4	2.4	2.4	+0.0 [−1.9, 1.9]
PHTest harmless	449	6.0	8.2	7.6	−0.7 [−2.7, 1.6]
Harmless request + transplanted suffix	100	1.0	1.0	2.0	+1.0 [0.0, 3.0]
First test, controversial, Refused %					
PHTest controversial	225	18.2	22.2	21.3	−0.9 [−4.4, 2.7]
Second test, harmful, Unsafe % ↓					
WildJailbreak, new requests	300	10.7	8.3	5.7	−2.7 [−5.7, 0.0]
In-the-wild templates	300	28.7	22.3	14.7	−7.7 [−12.0, −3.3]
MultiJail English	103	1.0	1.0	1.0	+0.0 [0.0, 0.0]
MultiJail, 9 languages	927	4.1	3.5	2.6	−0.9 [−2.0, 0.3]
MultiJail, high-resource (zh, it, vi)	309	2.3	2.3	1.9	−0.3 [−1.0, 0.0]
MultiJail, medium-resource (ar, ko, th)	309	4.9	2.9	3.6	+0.6 [−1.3, 2.9]
MultiJail, low-resource (bn, sw, jv)	309	5.2	5.2	2.3	−2.9 [−5.5, −0.6]
Aya red-teaming, six languages	900	12.2	10.9	7.8	−3.1 [−4.4, −1.7]
Aya red-teaming, Filipino	150	13.3	12.7	8.7	−4.0 [−8.7, 0.0]
Aya red-teaming, French	150	8.0	6.7	4.7	−2.0 [−5.3, 1.3]
Aya red-teaming, Hindi	150	10.0	8.7	6.0	−2.7 [−6.0, 0.7]
Aya red-teaming, Russian	150	22.0	20.7	16.7	−4.0 [−7.3, −1.3]
Aya red-teaming, Serbian	150	8.7	7.3	5.3	−2.0 [−4.7, 0.7]
Aya red-teaming, Spanish	150	11.3	9.3	5.3	−4.0 [−8.0, 0.0]
Aya red-teaming, Arabic and English	200	11.5	7.0	9.0	+2.0 [−0.5, 5.0]
Formats, all eight	640	18.9	13.4	13.6	+0.2 [−3.0, 3.3]
Four system-prompt formats	320	18.4	11.2	11.6	+0.3 [−3.8, 4.1]
Three multi-turn formats	240	17.5	14.6	15.8	+1.2 [−2.9, 5.4]
Aya red-teaming, English	100	18.0	12.0	16.0	+4.0 [0.0, 9.0]
Aya red-teaming, Arabic	100	5.0	2.0	2.0	+0.0 [−3.0, 3.0]
Format, short system prompt	80	16.2	8.8	8.8	+0.0 [−6.2, 6.2]
Format, long system prompt	80	21.2	10.0	16.2	+6.2 [0.0, 12.5]
Format, persona preamble	80	13.8	10.0	12.5	+2.5 [−3.8, 10.0]
Format, Portuguese preamble	80	22.5	16.2	8.8	−7.5 [−13.8, −2.5]
Format, after two harmless turns	80	20.0	15.0	18.8	+3.8 [−1.2, 10.0]
Format, after a refused turn	80	8.8	7.5	10.0	+2.5 [−2.5, 7.5]
Format, preamble and two turns	80	23.8	21.2	18.8	−2.5 [−10.0, 3.8]
Format, in-the-wild template as the system prompt	80	25.0	18.8	15.0	−3.8 [−11.2, 3.8]
Second test, harmless, Refused % ↓					
WildJailbreak, new harmless	300	9.3	12.3	10.0	−2.3 [−4.7, 0.0]
OR-Bench-hard	250	42.8	48.8	41.6	−7.2 [−10.8, −4.0]
Alpaca	250	0.8	1.2	0.8	−0.4 [−1.2, 0.0]
In-the-wild templates, harmless	150	10.7	15.3	16.7	+1.3 [−3.3, 5.3]
Aya prompts, English	40	0.0	0.0	0.0	+0.0 [0.0, 0.0]
Aya prompts, 9 languages	360	1.7	1.7	1.4	−0.3 [−1.4, 0.6]
Aya prompts, high-resource	120	1.7	1.7	1.7	+0.0 [0.0, 0.0]
Aya prompts, medium-resource	120	2.5	2.5	2.5	+0.0 [−2.5, 2.5]
Aya prompts, low-resource	120	0.8	0.8	0.0	−0.8 [−2.5, 0.0]
Aya prompts, six languages	360	0.8	1.1	0.8	−0.3 [−0.8, 0.0]

Set	n	Base	Control	AyaDial	AyaDial − Control (95% interval)
Aya prompts, Filipino	60	0.0	0.0	0.0	+0.0 [0.0, 0.0]
Aya prompts, French	60	0.0	0.0	0.0	+0.0 [0.0, 0.0]
Aya prompts, Hindi	60	0.0	0.0	0.0	+0.0 [0.0, 0.0]
Aya prompts, Russian	60	0.0	0.0	0.0	+0.0 [0.0, 0.0]
Aya prompts, Serbian	60	1.7	3.3	1.7	−1.7 [−5.0, 0.0]
Aya prompts, Spanish	60	3.3	3.3	3.3	+0.0 [0.0, 0.0]
Formats, all eight	960	20.8	23.6	20.7	−2.9 [−5.0, −1.0]
Four system-prompt formats	480	21.9	24.4	21.7	−2.7 [−5.4, −0.2]
Three multi-turn formats	360	14.7	17.5	13.9	−3.6 [−6.4, −1.1]
Format, short system prompt	120	15.0	19.2	13.3	−5.8 [−10.0, −1.7]
Format, long system prompt	120	41.7	43.3	42.5	−0.8 [−5.0, 2.5]
Format, persona preamble	120	16.7	16.7	17.5	+0.8 [−5.0, 6.7]
Format, Portuguese preamble	120	14.2	18.3	13.3	−5.0 [−9.2, −1.7]
Format, after two harmless turns	120	11.7	15.0	10.8	−4.2 [−8.3, −0.8]
Format, after a refused turn	120	24.2	25.8	23.3	−2.5 [−7.5, 2.5]
Format, preamble and two turns	120	8.3	11.7	7.5	−4.2 [−8.3, 0.0]
Format, in-the-wild template as the system prompt	120	35.0	39.2	37.5	−1.7 [−8.3, 5.0]